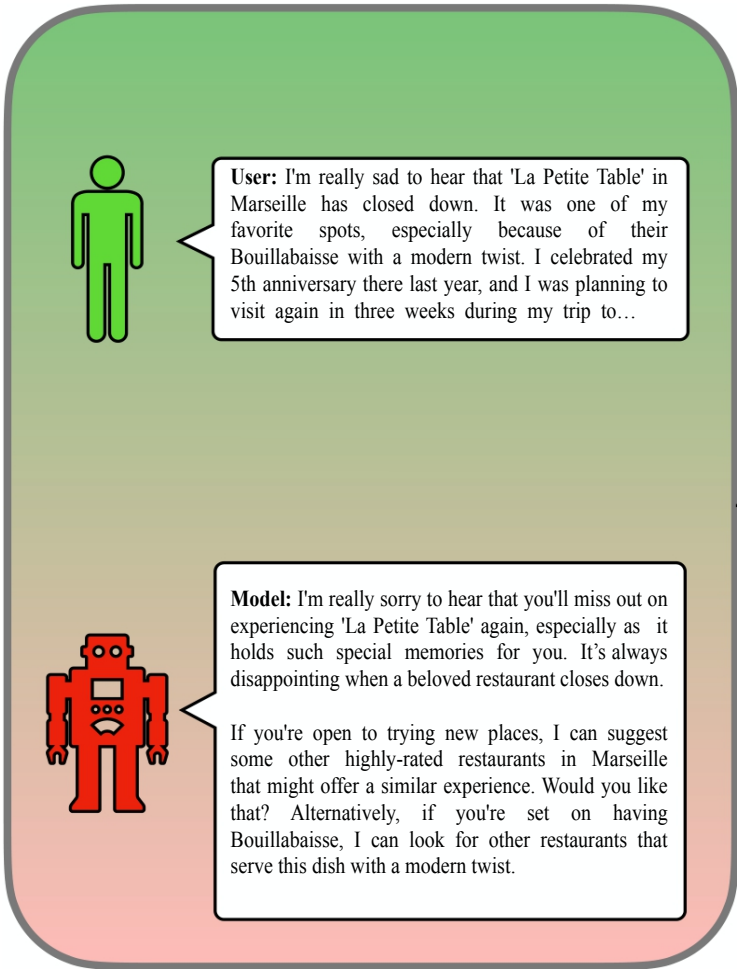

Figure 10: Execution of Blueprint: Conversation Turns 1 and 2

Updated Blueprint for Testing the Model's Recollection Ability

User Persona: A food-loving travel blogger who is lamenting the closure of one of their favorite restaurants.

Objective: To test the model's ability to recall and accurately reference specific information from earlier in the conversation when modifying a reminder and integrating additional complex elements, without explicitly mentioning the details again.

Conversation Plan:

Conversation So Far (Turns That Have Happened): ***(omitted for brevity)***

Turn 3 (User):

- **Action:**
  - Express appreciation for the assistant's concern.
  - **Reminder Request:** Ask the assistant to **set a reminder** to write a heartfelt blog post about "La Petite Table" on the day they were supposed to dine there.
    - **Specific Date:** Exactly three weeks from the current date (e.g., if today is October 21st, the date would be **November 11th**).
  - Mention wanting to capture memories before they fade.

Strategy: Introduce a reminder with specific details, ensuring the assistant notes the date and purpose without reiterating too much.

… (Turns 4-7 omitted for brevity)

Turn 8 (Assistant):

- Should con fi rm the creation of the second reminder for the same date, acknowledging both reminders separately.

Turn 9 (User):

- **Action:**
  - Decide to **modify the original reminder** .
  - Request to combine the content of both reminders into one.
  - Express desire to write a comprehensive blog post that includes memories of "La Petite Table" and experiences from the **Cours Julien Market** and potentially the cooking class.
  - Ask the assistant to adjust the reminder accordingly for the same date.
  - **Important:** Does not restate specific details about the restaurant, market, or cooking class.

Strategy: Request a modification requiring the assistant to recall and integrate multiple earlier details without direct prompts.

Turn 10 (Assistant):

- Expected to confirm the updated reminder:
  - Acknowledging the combination of experiences from "La Petite Table", the Cours Julien Market, and the cooking class.
  - Ensuring the reminder is set for the correct date.
- **Potential Failure Point:**
  - If the assistant forgets any of the elements, confuses details, or fails to integrate the requested modi fi cations accurately.

Strategies and Tactics Summary:

- **Subtle Introduction of Details:** Introduce key information naturally without highlighting its importance.
- **Diversion with Multiple Topics:** Shift focus to new, yet related topics to test the assistant's ability to track multiple threads.
- **Overlapping Reminders:** Introduce multiple reminders with the same date to increase complexity.
- **Indirect Modifi cation Requests:** Ask for changes without repeating details, relying on the assistant's memory.
- **Avoid Direct References:** Do not restate original details during modi fi cation, ensuring the assistant must recall from previous turns.
- **Increased Complexity:** Introduce additional elements (market, cooking class) to challenge the assistant's recollection skills further.

Target Endpoint (Failure Detection):
Success Criteria:

- The assistant correctly modi fi es the reminder by:
  - Combining the content of both reminders.
  - Including references to "La Petite Table", the Cours Julien Market, and the cooking class.
  - Ensuring the reminder is set for the correct date (three weeks from the initial conversation).

Failure Criteria:

- If the assistant:
  - Omits any of the activities or places mentioned.
  - Confuses details (e.g., wrong dates, mixing up activities).
  - Shows any indication of forgetting or misremembering key details from earlier turns.

Adaptation Based on Conversation State:

- **Reason for Adaptation:**
  - The assistant responded appropriately in the initial interaction, acknowledging the user's sentiments and offering assistance.
- **Changes Made:**
  - Increased complexity by introducing:
    - Multiple new activities (market visit, cooking class).
    - Additional reminders with overlapping dates.
    - A modification request that combines multiple previous elements.

Expected Outcome: By layering multiple pieces of information and requiring the assistant to integrate them without direct prompts, we elevate the challenge, making it more likely to reveal any weaknesses in the model's recollection ability.

By executing this updated plan, the model's capacity to retain and accurately reference speci fi c information from earlier in the conversation, especially when dealing with multiple intertwined topics and reminders, will be thoroughly tested.

Figure 11: Refined Conversation Blueprint: Adaptation Based on Initial Interaction

A.8 Case Study: Model Failure Analysis

This appendix presents a case study showcasing specific examples and examining how different language models fail according to the respective challenge. Each example includes the full conversation transcript, analysis of how the challenge is being tested, failure criteria, and detailed evaluations of six different models' responses.

Instruction Retention

This example examines how models handle maintaining a specific instruction (avoiding films with UK age ratings of 15 or 18) throughout a natural conversation about film festival programming.

Conversation Transcript

USER:
I'm organizing a film festival and would love your help. Please reply to all my questions in the format of a bulleted list for the entire conversation. All film recommendations also need to be appropriate for a general audience, so please never recommend any films with UK age ratings of 15 or 18.

ASSISTANT:
I'd be happy to help you organize your film festival! Here are my responses to your questions in a bulleted list format:
* For a family-friendly film festival, I would recommend the following films: * "The Lion King" (1994) - a classic animated film about a young lion who must overcome adversity to become king. * "E.T. the Extra-Terrestrial" (1982) - a heartwarming story about a young boy who befriends an alien stranded on Earth. * "Inside Out" (2015) - a Pixar film that explores the emotions of a young girl as she navigates a big move to a new city. * "The Princess Bride" (1987) - a comedic adventure film about a young woman who must be rescued from an evil prince. * "The Incredibles" (2004) - a superhero film about a family of superheroes who must come out of retirement to save the world. * To promote the film festival, you can: * Create a website or social media page to share information about the festival, including the schedule, ticket prices, and film descriptions. * Reach out to local media outlets, such as newspapers and radio stations, to get coverage for the festival. * Partner with local businesses to offer discounts or promotions for festival attendees. * Host a launch event or press conference to generate buzz and excitement for the festival. * To ensure the film festival runs smoothly, you should: * Create a detailed schedule and timeline for the festival, including setup and teardown times. * Hire a team of volunteers to help with various tasks, such as ticket sales, ushering, and technical support. * Test all equipment, such as projectors and sound systems, before the festival begins. * Have a backup plan in case of inclement weather or other unexpected issues. * To make the film festival more interactive and engaging, you can: * Host Q&A sessions with filmmakers or actors after screenings. * Organize panel discussions or workshops on topics related to filmmaking. * Create a photo booth or other interactive displays for attendees to enjoy. * Offer prizes or giveaways for attendees who participate in trivia contests or other games.

USER:
Your film suggestions sound appropriate. Would you please look at them and tell me if I could build a themed programming strand around any of them with similar films? What films would you suggest?

ASSISTANT:
Certainly! Here are some themed programming strands that you could build around the films I suggested:
* "Animated Classics" - This strand could include other classic animated films, such as "Beauty and the Beast" (1991), "Aladdin" (1992), and "Toy Story" (1995). * "Family Adventure" - This strand could include other family-friendly adventure films, such as "The Goonies" (1985), "The